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AI25BTECH11038 – Tejas Uppala

Question:

The scalar $\mathbf{A} \cdot ((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C}))$ equals

- | | |
|---------------------|--------------------------------------|
| 1) 0 | 3) $[\mathbf{ABC}] + [\mathbf{BCA}]$ |
| 2) $[\mathbf{ABC}]$ | 4) None of these |

Solution:

The expression is a scalar triple product, which can be written using the box product notation:

$$E = \mathbf{A} \cdot ((\mathbf{B} + \mathbf{C}) \times (\mathbf{A} + \mathbf{B} + \mathbf{C})) = [\mathbf{A} \quad (\mathbf{B} + \mathbf{C}) \quad (\mathbf{A} + \mathbf{B} + \mathbf{C})] \quad (1)$$

The box product is equivalent to the determinant of the matrix formed by the vectors. We can use determinant properties (where rows are the vectors $\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3$) to simplify:

Row Operation: Apply the determinant property that allows replacing one row by the sum/difference with other rows.

Operation: Replace the third vector ($\mathbf{R}_3$) by subtracting the first two vectors ($\mathbf{R}_1$ and $\mathbf{R}_2$):

$$\mathbf{R}'_3 = \mathbf{R}_3 - \mathbf{R}_1 - \mathbf{R}_2 \quad (2)$$

The new third vector is:

$$\mathbf{R}'_3 = (\mathbf{A} + \mathbf{B} + \mathbf{C}) - \mathbf{A} - (\mathbf{B} + \mathbf{C}) = \mathbf{A} + \mathbf{B} + \mathbf{C} - \mathbf{A} - \mathbf{B} - \mathbf{C} = \mathbf{0} \quad (3)$$

The box product becomes:

$$E = [\mathbf{A} \quad (\mathbf{B} + \mathbf{C}) \quad \mathbf{0}] \quad (4)$$

The determinant (or scalar triple product) is **zero** if one of the vectors is the zero vector ($\mathbf{0}$).

$$E = 0$$

The answer is (A) **0**